



Saint Columban College
College of Computing Studies
Pagadian City



Commitment Issues

Commitment Issues 



Easy General Skills picoCTF 2024 browser_webshell_solvable git

AUTHOR: JEFFERY JOHN

Hints 

Description

I accidentally wrote the flag down. Good thing I deleted it!

You download the challenge files here:

- [challenge.zip](#)

1 2 3

31,708 users solved



97%

Liked



 picoCTF{FLAG}

Submit

Author: The Analyst: Hyposelenia

Challenge: Commitment Issues

Category: General Skills

Date: 01/14/26



I. Objective

The objective of this challenge is to recover a deleted flag by analyzing the version history of a Git repository included in the challenge files.

II. Background

The challenge provides a ZIP file containing a directory with a hidden .git folder, indicating that the files are tracked using **Git version control**. Although the current version of the file does not contain the flag, Git preserves a complete history of changes made to tracked files. By inspecting previous commits, deleted or modified content can be recovered.

The challenge description and hints emphasize that version control systems allow recovery of earlier file states, even after sensitive information has been removed.

III. Tool Used

- **Oracle VirtualBox (Kali Linux)** – Used as the Linux environment
- **Linux Terminal** – Used to execute decoding commands
- **git - Version control system used for commit history analysis**

IV. Methodology

1. I used my Oracle VirtualBox – Kali linux for this challenge
2. I unzip the file after downloading it



```
(kali㉿kali)-[/media/sf_Downloaded_Cybersecfiles]
$ unzip challenge.zip
Archive:  challenge.zip
  creating: drop-in/
  creating: drop-in/.git/
  creating: drop-in/.git/branches/
  inflating: drop-in/.git/description
  creating: drop-in/.git/hooks/
  inflating: drop-in/.git/hooks/applypatch-msg.sample
  inflating: drop-in/.git/hooks/commit-msg.sample
  inflating: drop-in/.git/hooks/fsmonitor-watchman.sample
  inflating: drop-in/.git/hooks/post-update.sample
  inflating: drop-in/.git/hooks/pre-applypatch.sample
  inflating: drop-in/.git/hooks/pre-commit.sample
  inflating: drop-in/.git/hooks/pre-merge-commit.sample
  inflating: drop-in/.git/hooks/pre-push.sample
```

3. Navigated by changing my directory into the file extracted after unzipping

```
File Actions Edit View Help
  creating: drop-in/.git/logs/refs/heads/
  inflating: drop-in/.git/logs/refs/heads/master
  extracting: drop-in/message.txt
(kali㉿kali)-[/media/sf_Downloaded_Cybersecfiles]
$ cd drop-in
```

4. List files including **hidden** directories by using “ls -a”

```
(kali㉿kali)-[/media/sf_Downloaded_Cybersecfiles/drop-in]
$ ls
message.txt

(kali㉿kali)-[/media/sf_Downloaded_Cybersecfiles/drop-in]
$ ls -a
.  ..  .git  message.txt
```

This confirmed the presence of a .git directory, indicating a Git repository.



5. Check the current contents of the file by using “cat message.txt” as command. The file only contained the text TOP SECRET, suggesting the flag was removed.
6. View commit history by using “git log” command

```
(kali㉿kali)-[/media/sf_Downloaded_Cybersecfiles/drop-in]
$ git log
commit e1237df82d2e69f62dd53279abc1c8aeb66f6d64 (HEAD -> master)
Author: picoCTF <ops@picoctf.com>
Date: Sat Mar 9 21:10:14 2024 +0000

    remove sensitive info
Network
commit 3d5ec8a26ee7b092a1760fea18f384c35e435139
Author: picoCTF <ops@picoctf.com>
Date: Sat Mar 9 21:10:14 2024 +0000

    create flag
```

Two commits were found:

- One commit labeled “*remove sensitive info*”
- An earlier commit labeled “*create flag*”

7. Inspect the earlier commit using the command “git show <commit_hash>” (latest)”



```
(kali㉿kali)-[/media/sf_Downloaded_Cybersecfiles/drop-in]
$ git show e1237df8

commit e1237df82d2e69f62dd53279abc1c8aeb66f6d64 (HEAD → master)
Author: picoCTF <ops@picoctf.com>
Date: Sat Mar 9 21:10:14 2024 +0000

    remove sensitive info

diff --git a/message.txt b/message.txt
index 96f7309..d552d1e 100644
--- a/message.txt
+++ b/message.txt
@@ -1,1 @@
-picoCTF{s@n1t1z3_30e86d36}
+TOP SECRET

(kali㉿kali)-[/media/sf_Downloaded_Cybersecfiles/drop-in]
$
```

Viewing the earlier commit revealed the original contents of message.txt, including the flag.

V. Result

picoCTF{s@n1t1z3_30e86d36}

```
@@ -1,1 @@
-picoCTF{s@n1t1z3_30e86d36}
+TOP SECRET
```

VI. Explanation

Although the flag was deleted in the most recent commit, Git retains snapshots of all previous versions of tracked files. The git log command was used to identify an earlier commit where the flag still existed. Using



git show allowed inspection of the file contents at that point in time, effectively recovering the deleted flag.

This demonstrates how version control systems can unintentionally expose sensitive information if repository history is not properly sanitized.

VII. Conclusion

This challenge highlights the forensic value of Git commit history. Even when files appear clean in the current version, previously committed sensitive data can often be recovered. Understanding Git commands such as git log, git show, and commit checkout is essential for both secure development practices and cybersecurity investigations.

— **The Analyst: Hyposelenia**